ONNX Remediation | Engineering Gate v0.3

Smart Pet Vision - AI251P Applied Computer Vision | Phase 2 / M8 | 31 October 2026

Release status: NOT RELEASE-READY | Original project tolerance: 0.001 (absolute raw-head maximum). No model weights, Phase 1 assets or GitHub main branch were modified.

1. Verified inputs and provenance

Artifact / gate	Verified observation
Frozen Drive source	30 of 30 SHA256 checks passed
Detector PyTorch checkpoint	best.pt - SHA256 f48e4d8e...125da1cd4
Detector baseline ONNX export	best.onnx - SHA256 e8b55114...34862
Parity cohort	50 authenticated files: 25 multi-animal, 12 cat-single, 13 dog-single
Private data handling	No original team photos or model binaries included in the release ZIP

2. Acceptance evidence

Check	Result	Interpretation
Historical raw-head parity	0.00201416 > 0.001	FAIL (existing measurement)
Historical 3-image final boxes	0.00012207 px	Narrow post-NMS result; does not override raw FAIL
New ONNX-specific unit tests	16 / 16 passed	Software logic only
Full source test suite	62 passed, 3 import errors	Model library unavailable
Live model parity attempt	BLOCKED: onnxruntime missing	Not a new parity measurement

3. Engineering changes (candidate, not deployed)

• New fail-closed parity policy: every test must preserve the 0.001 raw-head threshold, use at least 50 real images, and compare final counts/classes/boxes as a separate condition.

• New read-only diagnostic matrix: ONNX Runtime graph optimization and PyTorch MKLDNN modes; optional independent ONNX re-export without overwriting the baseline.

• After-NMS confidence is compared as well as boxes and classes; inconsistent or empty predictions cannot become a false PASS.

• Fixed a pre-existing unit test that could report success after a premature hash-key error. Correct keys now verify the approved model SHA values before blocking.

• Created an executable Colab verification notebook, with explicit Drive inputs, frozen parity sample, dependency setup and machine-readable PASS / FAIL / BLOCKED logs.

4. Final release decision

ONNX: NO-GO / keep PyTorch backend. A numerical-fix claim is impossible without running the real checkpoint in a compatible environment. When a reproducible candidate passes raw and final parity, separately evaluate the complete current pipeline, multi-animal referral policy and T4 runtime. Until then, every technical Ready image still requires human approval.

Conditional publication: Do not push a final artifact to Drive or GitHub until real-model parity is verified, source discrepancies are reconciled, and the project gate is signed off. Only a separate Phase 2 branch/PR is permitted; never rewrite submitted Phase 1.